

**BUSINESS REQUIREMENTS
& PROJECT SCOPE DOCUMENT**

Pet Services Mobile Application

Saudi Arabia

Flutter · Firebase · Arabic RTL + English

Version 1.0 (First Draft)

Prepared for development handoff

10-Week Engagement · 3 Phases · Fixed Price

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1. Executive Summary

This document defines the complete business and technical requirements for a mobile pet-services application targeting Saudi Arabia. The application connects pet owners with on-demand pet transport, pet-care service providers, animal sellers, and emergency veterinary clinics, and includes a community rescue-map feature for stray and distressed animals.

The core of the product is the driver/transport service. Every other feature — services, housing, marketplace, rescue, clinic finder — connects to and is supported by the transport layer. The application is bilingual (Arabic RTL and English LTR) and is delivered as a Flutter mobile app with a Firebase-managed backend.

Project Snapshot

Item	Detail
Product	Pet-services mobile application (iOS + Android)
Frontend	Flutter (single codebase, iOS + Android)
Backend	Firebase + managed SQL database (no self-hosted servers)
Languages	Arabic (RTL) and English (LTR)
Total screens	60 (status to be re-verified; treat all as requiring design/build confirmation)
Markets	Saudi Arabia
Duration	10 weeks across 3 phases
Engagement	Fixed price USD 2,000 — contractor only
Ownership	Client owns 100% of code, IP, data, and assets

Engagement note: The developer is engaged as an independent contractor. The developer is not a co-founder, partner, shareholder, or stakeholder, and acquires no ownership, equity, or revenue rights. See Section 17.

2. Project Objectives

The project must deliver, in priority order:

- **A working pet-transport app, where a user can request transport for a pet from any pickup point to any destination (registered or custom).**
- **A pet-care services marketplace (grooming, treatment, hiking, showering) with provider discovery, booking, scheduling, and payment.**
- **A pet housing marketplace covering both formal providers (clinics, stores) and informal home-based hosts.**
- **A buy/sell/adopt animals marketplace, including a verified seller tier for licensed/high-value animals.**
- **An emergency 24/7 vet-clinic finder with map, filtering, and pay-to-book.**
- **A community rescue map where any user can pin a distressed animal and others can respond.**
- **A pet medical-supplies request system: user requests are received and manually fulfilled by admin.**
- **Controlled in-app chat scoped strictly to completing a transaction.**

Success Criteria

- A user can register, add a pet, book a transport request, pay, and track the status of the request through to completion without errors.
- A service provider can register, get verified, list services, receive bookings, and view earnings.
- All 60 screens are implemented in both Arabic RTL and English LTR.
- All Firebase resources, source code, and credentials are transferred to the client at handover.
- The codebase is maintainable by any competent Flutter developer without the original developer.

3. Scope

3.1 In Scope

- Flutter mobile application for iOS and Android
- Backend: Firebase Authentication, Storage, Cloud Functions, Hosting, Analytics, Crashlytics, Cloud Messaging, Security Rules — plus a managed SQL database as the main datastore (Firestore used as secondary storage where it fits better, e.g. listings, chat, rescue pins)
- All Figma screens completed, plus Flutter implementation of all 60 screens
- Arabic RTL and English LTR support across the entire app
- Payment gateway integration (Moyasar or Tap, supporting Mada and card)
- Push notifications for ride/booking status, chat, and verification events
- A mobile-based admin app for user management, verification approvals, driver/supply request allocation, content moderation, and reports
- Full documentation, testing, and handover deliverables

3.2 Out of Scope (V2 / Future)

- Remote vet video-consultation sessions
- Live tracking of pets/drivers on the map
- Driver registration (Uber-style self-service driver onboarding)
- In-app pet medical records
- Prescription delivery
- Group chat per rescue case and rescue volunteer profiles
- Loyalty / referral programs
- Web or desktop consumer app (mobile only for this engagement)
- Advanced analytics dashboards beyond the reports defined in Section 12

Scope control: Any feature not explicitly listed in Section 3.1 is out of scope for the USD 2,000 / 10-week engagement and is subject to a separate agreement.

4. User Roles

The application defines the following roles. A single account may hold more than one role (for example, a pet owner who is also a home houser). Role is stored on the user record and enforced by Security Rules (Firestore) and access checks (SQL) in Cloud Functions.

Role	Description	Verification Required
Pet Owner / Buyer	Default role. Books transport, services, housing; buys/adopts animals; buys supplies; uses rescue map and clinic finder.	No (phone OTP only)
Rescuer	Not a separate account type. Any Pet Owner can pin a rescue case or respond to one.	No
Seller	Lists animals for sale or adoption; lists medical supplies.	No for standard listings
Verified / Licensed Seller	Lists high-value/regulated animals (hawks, trained dogs, exotics). Receives a verified badge.	Yes (documents)
Service Provider	Offers grooming, treatment, hiking, showering. Manages bookings and earnings.	Yes (documents)
Home Houser	Offers home-based pet boarding. Manages availability and bookings.	Yes (documents)
Pet Clinic	Veterinary clinic. Listed in clinic finder; flagged if 24/7; can receive pay-to-book.	Yes (documents)
Admin	Platform staff. Approves verifications, allocates drivers and supply requests, moderates content, views reports. Mobile admin app only.	Internal account

Pet transport drivers are not app accounts in this MVP. Drivers are coordinated manually by admin (outside the formal role model above) and are allocated to ride requests through the admin app — see Section 7.

4.1 Permission Matrix (summary)

Action	Owner	Seller	Provider	Admin
Book a ride / service	Yes	Yes	Yes	No
Allocate a driver to a ride	No	No	No	Yes
Create animal listing	No	Yes	No	Yes
Create service listing	No	No	Yes	No
Approve verification	No	No	No	Yes
Pin rescue case	Yes	Yes	Yes	Yes
View all users	No	No	No	Yes
Moderate/close chat	No	No	No	Yes
View own earnings	No	Yes	Yes	Yes

5. Functional Requirements

Each functional module below specifies frontend requirements, backend requirements, Firestore collections used, security/validation, and notifications. Page-level detail is in Section 6; data structures are in Section 8.

5.1 Authentication & Account

Frontend

- Phone-number + password login; Google Sign-In; Skip-for-now guest browsing
- Sign-up: username, phone, password; phone verified by OTP
- Forgot-password flow: phone → OTP → set new password
- Account-type selection panel; document upload for roles requiring verification

Backend

- Firebase Authentication with Phone provider and Google provider
- On first sign-up, a Cloud Function creates the user document in Firestore with default role `pet_owner`
- OTP delivered via Firebase Phone Auth; rate-limited (max 3 requests / 10 min / number)

Data Stored

- users, verificationRequests

Validation & Security

- Phone must be E.164 format; password minimum 8 chars; username 3–30 chars
- A user can read/write only their own user document; admin can read all

Notifications

- OTP via SMS (Firebase); welcome in-app notification on first login

5.2 Pet Transport (Core)

Frontend

- Ride request: pickup (search + map pin), destination (search + map pin, any location), select pet, car type (Any/Specific)
- Route + fare preview on map before confirmation
- Live driver tracking on map with driver card (name, vehicle, plate, ETA)
- Safety compliance screen: OTP, live animal photo, cage-closed confirmation, location confirmation
- Ride status: Requested → Driver Allocated → In Progress → Delivered / Cancelled

Backend

- On ride request, a Cloud Function creates a ride record with status requested; the request appears in the admin app's Pending Driver Requests queue
- Admin manually allocates a driver to the request via the admin app; ride status updates to `driver_allocated` and the user is notified
- Fare computed by a Cloud Function from distance, time, and surge; never trusted from the client

Data Stored

- rides (SQL — main record), drivers (SQL — simple roster: name, phone, vehicle, plate, active flag)

Validation & Security

- Only admin can change a ride's status or assign a driver
- Only the owning user can cancel their own ride (before driver allocation)
- Fare amount is written only by the Cloud Function (Admin SDK), never client-side

Notifications

- Push to user when admin allocates a driver; push to user on status changes and delivery

5.3 Pet-Care Services

Frontend

- Provider discovery (list + map); provider profile with services and reviews
- Select pet → choose service → calendar/time-slot → payment → confirmation
- Post-booking prompt: 'Request a driver for transport?'

Backend

- Bookings created; slot availability checked by a Cloud Function transaction to prevent double-booking a slot
- Payment captured via gateway Cloud Function before booking is confirmed

Data Stored

- services, bookings, providers, reviews

Validation & Security

- A booking references a valid serviceId, providerId, petId owned by the user, and an available slot
- Only the provider or the booking owner can read a booking; provider can update status

Notifications

- Push to provider on new booking; push to user on confirmation and reminders

5.4 Pet Housing

Frontend

- Search (list + map), filter (Home Care / Clinic Stay / 24-7 Med), listing detail with photos, price/night, availability calendar, reviews
- Booking: dates → pet → price breakdown → pay → confirmation; driver drop-off prompt

Backend

- Housing listings created; availability stored per listing; booking blocks dates via Cloud Function transaction

Data Stored

- housingListings, housingBookings, reviews

Validation & Security

- Booking dates must not overlap an existing booking for the same listing
- Houser can read bookings for their listings; owner can read their own bookings

Notifications

- Push to houser on new booking; push to user on confirmation

5.5 Buy / Sell / Adopt Animals

Frontend

- Listing page (search + filter by type), animal detail (photos, details, price, Flag Interest, Provider Details, Ask About Pet, Buy)
- Adoption listings identical but with no price ('Adopt Free')
- Verified seller section with verified badge for licensed/high-value animals

- Seller: create listing, manage listings (edit / delete / mark sold)

Backend

- Listings created; images in Storage; verified flag set only by admin after approval
- Purchase creates an order and triggers payment Cloud Function

Data Stored

- animalListings, orders, verificationRequests

Validation & Security

- Seller can edit/delete only their own listings; verified flag writable only by admin
- Adoption listings must have price == null; sale listings must have price > 0

Notifications

- Push to seller on interest flag and on purchase; in-app to buyer on order status

5.6 Emergency 24/7 Clinic Finder

Frontend

- Clinic list + map; filters: All / Open Now / 24-7 / Nearest; clinic card (name, distance, open status, hours, call)
- Clinic detail (hours, address, rating, Call, Book, Get Driver)
- Pay-to-book: pay now to secure a slot, with chat access until visit

Backend

- Clinics (admin-maintained) with an is24Hours flag and opening hours; 'Open Now' computed client-side from hours + device time
- Pay-to-book creates a clinicBooking and captures payment via Cloud Function

Data Stored

- clinics, clinicBookings

Validation & Security

- is24Hours and clinic data writable only by admin
- clinicBooking readable by the owning user and the clinic

Notifications

- Push/in-app confirmation on pay-to-book; reminder before visit

5.7 Rescue Map

Frontend

- Map with animal pins; active-pin banner; pin detail bottom sheet with 'I Can Help'
- Drop pin: animal type, urgency (Critical/Needs Help/Stable), optional short text, location
- No photo upload required (storage-cost control)

Backend

- Rescue pins (Firestore) with a GeoPoint; nearby query via geohash field; pins auto-expire (status closed) after a configurable period
- 'I Can Help' records a responder on the pin and notifies the reporter

Data Stored

- rescuePins

Validation & Security

- Any authenticated user can create a pin; only the reporter or admin can close it
- Optional text limited to 200 chars; no contact info (filtered)

Notifications

- Push to nearby users on a new Critical pin; push to reporter when someone offers help

5.8 Pet Medical Supplies (Manual Fulfillment)

Frontend

- Request page (category filter): user submits a request for a supply item (medication, cage, equipment); item detail shown when available
- Create request: photo (optional), title, category, description

Backend

- Supply requests created; admin reviews via Pending Medication/Supply Requests in the admin app and sets status to approved, completed, or unavailable

Data Stored

- supplyRequests, orders

Validation & Security

- Requester can view only their own requests; status change restricted to admin; medication category requires admin sign-off before marking approved

Notifications

- Push to user when admin updates request status (approved / completed / unavailable)

5.9 Controlled Chat

Frontend

- Chat history (one thread per transaction); chat screen with monitored banner, text only, no media button
- Entry points: Ask About Pet, Ask Seller, provider/houser contact, clinic pay-to-book

Backend

- Threads and messages (Firestore); a Cloud Function (or client-side regex pre-check + server validation) blocks phone numbers, emails, and URLs
- Thread auto-locks when the related transaction is completed or cancelled

Data Stored

- chatThreads, chatThreads/{id}/messages

Validation & Security

- Only the two participants (and admin) can read a thread; messages immutable once sent
- Messages matching blocked patterns are rejected with an error to the sender

Notifications

- Push to recipient on a new message while thread is active

6. Page-by-Page Specifications

All 60 screens are specified below, grouped by module. Each screen lists purpose, access, entry point, layout, components, states, responsive behavior, and Firebase/SQL interactions. Status: [E] = exists in Figma, [N] = new (design + build), [P] = partial (exists, needs completion). Note: all statuses should be treated as requiring re-verification (see Section 1).

Authentication & Profile

6.1 Login Page [E]

Attribute	Specification
Purpose	Entry point; authenticate via phone+password or Google, or skip to browse.
Access	Guest / unauthenticated
Entry point	App launch (if not logged in)
Layout	Logo top; phone field; password field; Sign In button; Forgot Password link; Sign In with Google; Sign Up link; Skip For Now.
Components	2 inputs, primary button, Google button, 2 text links
States	Empty (default), Loading (on submit), Error (invalid credentials banner), Success (navigate Home)
Mobile/Tablet/Desktop	Single-column centered; max width 420px on tablet/desktop
Firebase	Firebase Auth signInWithPhone / signInWithCredential (Google)

6.2 OTP Confirmation [E]

Attribute	Specification
Purpose	Verify phone via one-time code.
Access	Guest (during sign-up / reset)
Entry point	After Sign Up or Forgot Password
Layout	Phone display; OTP input (6 boxes); Send/Resend with countdown; Confirm button.
States	Empty, Loading, Error (wrong/expired code), Success (proceed)
Firebase	Firebase Phone Auth verify; rate-limited

6.3 Set Password [E]

Attribute	Specification
Purpose	Set a new password after OTP (reset flow).
Access	Guest (post-OTP)
Entry point	After OTP in reset flow
Layout	New Password; Confirm Password; Change Pass button.
States	Empty, Error (mismatch / weak), Success
Firebase	Auth updatePassword

6.4 Register Page [E]

Attribute	Specification
Purpose	Create a new account.
Access	Guest
Entry point	Sign Up link on Login
Layout	Username, phone, password, city/address; Submit.
States	Empty, Validation errors per field, Loading, Success → OTP
Firebase	Auth createUser + Cloud Function creates users doc

6.5 Sign In With Google [E]

Attribute	Specification
Purpose	OAuth consent.

Attribute	Specification
Access	Guest
Entry point	Google button on Login
Layout	Google branding; access description.
Firebase	Google provider credential

6.6 Forgot Password [N]

Attribute	Specification
Purpose	Start password reset by phone.
Access	Guest
Entry point	Forgot Password link on Login
Layout	Phone field; Send OTP; Back to Login.
States	Empty, Error (no account), Loading, Success → OTP
Firebase	Auth phone verify

6.7 Display User Info [N]

Attribute	Specification
Purpose	Read-only profile view.
Access	Authenticated owner
Entry point	Side Bar → User Settings
Layout	Avatar, name, phone, city, email; Edit button.
States	Loaded, Loading skeleton
Firebase	Read users/{uid}

6.8 Edit User Info [N]

Attribute	Specification
Purpose	Edit profile details.
Access	Authenticated owner
Entry point	Display User Info → Edit
Layout	Avatar upload; name, phone, city, email; Save.
States	Editing, Saving, Error, Success
Firebase	Update users/{uid}; avatar → Storage

6.9 My Pets List [N]

Attribute	Specification
Purpose	List user's pets.
Access	Authenticated owner
Entry point	Side Bar → Your Pets
Layout	Pet cards (photo, name, species, age); Add New Pet card.
States	Empty ('No pets yet — add one'), Loaded, Loading
Firebase	Query pets where ownerId == uid

6.10 Add Pet [N]

Attribute	Specification
Purpose	Register a new pet.
Access	Authenticated owner
Entry point	My Pets → Add
Layout	Photo upload; name, species, breed, age, gender toggle; Save.

Attribute	Specification
States	Empty, Validation, Saving, Success
Firebase	Create pets doc; photo → Storage

6.11 Edit Pet Info [N]

Attribute	Specification
Purpose	Edit or delete a pet.
Access	Authenticated owner
Entry point	My Pets → tap pet
Layout	Pre-filled Add Pet form; Delete option.
States	Editing, Saving, Deleting, Success
Firebase	Update / delete pets/{id}

6.12 Service History [N]

Attribute	Specification
Purpose	List past orders/rides/bookings.
Access	Authenticated owner
Entry point	Side Bar → Orders
Layout	List of order cards (type, date, status badge, price).
States	Empty, Loaded, Loading, Filter by type
Firebase	Query rides/bookings/orders where userId == uid

Home & Navigation

6.13 Home Page [E]

Attribute	Specification
Purpose	Main dashboard with quick actions.
Access	Authenticated (guest limited)
Entry point	Post-login landing
Layout	Greeting; service tiles (Grooming, Shower, Delivery); Buy/Sell section; emergency banner if pins active; + FAB.
States	Loaded, Loading, Guest (some actions prompt login)
Firebase	Read featured listings; rescuePins count

6.14 Side Bar [E]

Attribute	Specification
Purpose	Navigation drawer.
Access	Authenticated
Entry point	Hamburger / + on Home
Layout	User Settings, Your Pets, Orders, Account Type, Payment Methods.
Firebase	Read users/{uid} for header

Controlled Chat

6.15 Chat History [N]

Attribute	Specification
Purpose	List active transaction threads.

Attribute	Specification
Access	Authenticated participant
Entry point	Bottom nav / from listings
Layout	Thread rows (name, last message, time, unread dot).
States	Empty, Loaded, Loading
Firebase	Query chatThreads where participants contains uid

6.16 Chat Screen [N]

Attribute	Specification
Purpose	Text-only transaction messaging.
Access	Two participants + admin
Entry point	Chat History / Ask About Pet / Ask Seller
Layout	Monitored banner; message bubbles; text input + send (no attachment).
States	Active, Locked (transaction done), Blocked-message error
Firebase	Listen messages subcollection; create message (validated)

Buy / Adopt Animals

6.17 Search / Listing Page [E]

Attribute	Specification
Purpose	Browse animals for sale.
Access	Authenticated / guest
Entry point	Buy Pets tab
Layout	Search bar; filter button; animal cards (photo, info, price); infinite scroll.
States	Empty, Loaded, Loading, Paginating
Firebase	Query animalListings paginated

6.18 Filter Page [E]

Attribute	Specification
Purpose	Narrow results.
Access	Authenticated / guest
Entry point	Filter button
Layout	Category chips (Cats, Dogs, Horses, Camel, Birds, Adoption); Apply.
Firebase	Re-query with filters

6.19 Animal Detail Page [E]

Attribute	Specification
Purpose	Full listing detail.
Access	Authenticated / guest
Entry point	Tap a listing
Layout	Photos; details; price; Flag Interest; Provider Details; Ask About Pet; Buy.
States	Loaded, Loading, Sold badge
Firebase	Read animalListings/{id}; create chat thread / order

6.20 Ask About Pet (Chat Entry) [E]

Attribute	Specification
Purpose	Open a controlled chat about a listing.

Attribute	Specification
Access	Authenticated buyer
Entry point	Animal Detail
Layout	Prompt + text input; Send.
Firebase	Create chatThread (buyer+seller)

6.21 Seller / Provider Details [E]

Attribute	Specification
Purpose	View seller/provider profile.
Access	Authenticated / guest
Entry point	Provider Details button
Layout	Image, name, address, listings/services.
Firebase	Read users/{sellerId} + their listings

6.22 Verified Sellers Section [N]

Attribute	Specification
Purpose	Curated licensed/high-value listings.
Access	Authenticated / guest
Entry point	Section/filter in Buy Pets
Layout	Verified badge; filtered listings (hawks, trained dogs, exotics).
Firebase	Query animalListings where verified == true

6.23 Adoption Listing Page [N]

Attribute	Specification
Purpose	Browse free-adoption animals.
Access	Authenticated / guest
Entry point	Adoption tab
Layout	Same as listing; 'Adopt Free' instead of price.
Firebase	Query animalListings where price == null

Checkout (Shared)

6.24 Checkout — Address & Payment [N]

Attribute	Specification
Purpose	Shared checkout for purchases/bookings.
Access	Authenticated
Entry point	Buy / Book / Pay actions
Layout	Address (search + map pin); payment method selector (Mada default); order summary; total; Confirm & Pay.
States	Empty, Validation, Processing, Success → Payment Success, Failure → Payment Failure
Firebase	Payment Cloud Function; create order/booking

6.25 Payment Success [N]

Attribute	Specification
Purpose	Confirm successful payment.
Access	Authenticated
Entry point	After successful payment
Layout	Green check; amount; order number; ETA; Track / Home.

Attribute	Specification
Firebase	Read created order

6.26 Payment Failure [N]

Attribute	Specification
Purpose	Handle failed payment.
Access	Authenticated
Entry point	After failed payment
Layout	Red X; reason; Try Again; Change Method.
Firebase	Retry payment function

Animal Services

6.27 Search for Service Provider [E]

Attribute	Specification
Purpose	Find providers (list + map).
Access	Authenticated
Entry point	Services tab / Home tile
Layout	Search; list/map toggle; provider cards (distance, rating).
States	Empty, Loaded, Loading, Map view
Firebase	Query providers (geo)

6.28 Setup Animal (Service) [E]

Attribute	Specification
Purpose	Choose/add pet before booking.
Access	Authenticated
Entry point	After choosing a service
Layout	Pets list (select) + Add New; Continue.
Firebase	Read pets; optional create

6.29 Service Provider Profile [E]

Attribute	Specification
Purpose	Provider services & availability.
Access	Authenticated
Entry point	Tap provider
Layout	Image, name, address, services, reviews.
Firebase	Read providers/{id}, services, reviews

6.30 Calendar / Time Slot [P]

Attribute	Specification
Purpose	Pick date & slot.
Access	Authenticated
Entry point	Provider profile → Book
Layout	Month calendar; available slots; Confirm Slot.
States	Loaded, No slots, Slot taken (refresh)
Firebase	Read availability; transaction on book

6.31 Select Payment (Service) [E]

Attribute	Specification
Purpose	Choose payment method.
Access	Authenticated
Entry point	Service booking flow
Layout	Methods list; Add New; Done.
Firebase	Read paymentMethods

6.32 Service Order Summary [N]

Attribute	Specification
Purpose	Review before confirm.
Access	Authenticated
Entry point	After slot + payment
Layout	Animal, service, time, provider, price; Confirm.
Firebase	Prepare booking

6.33 Service Order Placed [E]

Attribute	Specification
Purpose	Booking confirmation.
Access	Authenticated
Entry point	After confirm
Layout	Order details; 'Request Pet Delivery?' Yes/No.
Firebase	Create booking; optional ride

Driver / Transport

6.34 Ride Request Page [E]

Attribute	Specification
Purpose	Request pet transport.
Access	Authenticated
Entry point	Home Delivery tile / post-booking
Layout	Pickup (search+map); destination (search+map); select pet; car type; Continue.
States	Empty, Validation, Loading
Firebase	Create ride (requested)

6.35 Delivery Path Review [E]

Attribute	Specification
Purpose	Show route + fare.
Access	Authenticated
Entry point	After ride request
Layout	Map route; pickup/destination; price; Continue.
Firebase	Fare from Cloud Function

6.36 Setup Payment (Ride) [E]

Attribute	Specification
Purpose	Choose payment.
Access	Authenticated
Entry point	Ride flow

Attribute	Specification
Layout	Methods list; Add New; Done.
Firebase	Read paymentMethods

6.37 Searching For a Driver [E]

Attribute	Specification
Purpose	Wait for admin to allocate a specific driver.
Access	Authenticated
Entry point	After payment setup
Layout	Waiting state with status message; request summary; Cancel.
States	Pending allocation, Allocated (advance), Cancelled
Firebase	Listen ride status

6.38 Wait For Driver [E]

Attribute	Specification
Purpose	Admin has allocated a driver — details + status.
Access	Authenticated
Entry point	On allocation
Layout	Driver name, vehicle, plate; order details; status; Cancel.
Firebase	Read ride record

6.39 Ride Status [N]

Attribute	Specification
Purpose	Track the status of an active ride.
Access	User + Admin
Entry point	During ride
Layout	Driver card; status steps; Cancel.
States	Driver Allocated / In Progress / Delivered / Cancelled
Firebase	Listen ride record

6.40 Delivery Confirmation [E]

Attribute	Specification
Purpose	Confirm delivery complete.
Access	User + Admin
Entry point	On arrival
Layout	Confirm.
Firebase	Update ride status delivered

Emergency Clinic

6.41 24/7 Clinic Finder [N]

Attribute	Specification
Purpose	Find vet clinics, filter open/24-7.
Access	Authenticated / guest
Entry point	Home / emergency entry
Layout	Search; filter chips (All/Open Now/24-7/Nearest); clinic cards; map toggle.
States	Empty, Loaded, Loading, Map

Attribute	Specification
Firebase	Query clinics (geo); open-now computed client-side

6.42 Clinic Detail [N]

Attribute	Specification
Purpose	Clinic profile + actions.
Access	Authenticated / guest
Entry point	Tap clinic
Layout	Name, hours, address, rating; Call / Book / Get Driver.
Firebase	Read clinics/{id}

6.43 Pay-to-Book Emergency Slot [N]

Attribute	Specification
Purpose	Pay now to secure a slot + chat.
Access	Authenticated
Entry point	Clinic Detail → Book
Layout	Clinic summary; explainer; fee breakdown; Pay & Secure.
States	Empty, Processing, Success, Failure
Firebase	Create clinicBooking; payment function; open chat

Rescue Map

6.44 Rescue Map [N]

Attribute	Specification
Purpose	View distressed-animal pins.
Access	Authenticated
Entry point	Tab / Home banner
Layout	Map pins; active-pin banner; pin bottom sheet (I Can Help / Send Driver).
States	No pins, Pins loaded, Loading
Firebase	Query rescuePins (geo, status open)

6.45 Drop a Rescue Pin [N]

Attribute	Specification
Purpose	Report an animal in distress.
Access	Authenticated
Entry point	Rescue Map → + Pin
Layout	Animal type tiles; urgency; optional text; auto-GPS; Drop Pin.
States	Empty, Submitting, Success
Firebase	Create rescuePins (GeoPoint + geohash)

Housing

6.46 Housing Search / Listing [N]

Attribute	Specification
Purpose	Browse boarding options.
Access	Authenticated

Attribute	Specification
Entry point	Housing entry / Home
Layout	Search; filter (Home Care/Clinic/24-7 Med); cards (photo, price/night, rating); map.
States	Empty, Loaded, Loading, Map
Firebase	Query housingListings (geo)

6.47 Housing Detail [N]

Attribute	Specification
Purpose	Listing detail + availability.
Access	Authenticated
Entry point	Tap listing
Layout	Photos; price/night; accepted animals; availability calendar; reviews; Book.
Firebase	Read housingListings/{id}

6.48 Housing Booking Flow [N]

Attribute	Specification
Purpose	Book dates + pay.
Access	Authenticated
Entry point	Housing Detail → Book
Layout	Date picker; nights; pet; price breakdown; Confirm & Pay; driver prompt.
States	Empty, Dates conflict, Processing, Success
Firebase	Transaction blocks dates; payment function

Pet Medical Supplies

6.49 Supplies Request / Search [N]

Attribute	Specification
Purpose	Browse available supplies; submit a request.
Access	Authenticated
Entry point	Supplies entry
Layout	Search; category chips; request cards (photo, status).
Firebase	Query supplyRequests (own)

6.50 Supplies Request Detail [N]

Attribute	Specification
Purpose	View request detail and status.
Access	Authenticated
Entry point	Tap request
Layout	Photo (optional); description; status (Pending / Approved / Completed / Unavailable).
Firebase	Read supplyRequests/{id}

6.51 Create Supplies Request [N]

Attribute	Specification
Purpose	Submit a request for a supply item.
Access	Authenticated
Entry point	Supplies → Request
Layout	Photo (optional); title; category; description; Submit.

Attribute	Specification
States	Empty, Validation, Saving, Success
Firebase	Create supplyRequests; photo → Storage

Account & Verification

6.52 Account Type Selection [P]

Attribute	Specification
Purpose	Choose/change role.
Access	Authenticated
Entry point	Side Bar → Account Type
Layout	Options: User, Pet Store, Pet Clinic (add Seller, Home Houser); Confirm.
Firebase	Update users.role / roleRequest

6.53 Document Upload [E]

Attribute	Specification
Purpose	Submit verification docs.
Access	Roles needing verification
Entry point	After role selection
Layout	Upload area; required docs; Submit.
States	Empty, Uploading, Submitted
Firebase	Files → Storage; create verificationRequests

6.54 Verification Status [N]

Attribute	Specification
Purpose	Show approval status.
Access	Applicant
Entry point	After submit / Side Bar
Layout	Status icon; step progress; SMS note.
States	Pending, Approved, Rejected (with reason)
Firebase	Read verificationRequests/{id}

Service Provider & Seller Management

6.55 Available Appointments [N]

Attribute	Specification
Purpose	Provider's bookings.
Access	Service Provider
Entry point	Provider dashboard
Layout	Booking list (time, customer, service); Confirm/Reject.
Firebase	Query bookings where providerId == uid

6.56 Service Listings Mgmt [N]

Attribute	Specification
Purpose	Manage own services.
Access	Service Provider

Attribute	Specification
Entry point	Provider dashboard
Layout	Service list (name, price, duration); Edit/Delete; Add Service.
Firebase	CRUD services where providerId == uid

6.57 Earnings / Dashboard [N]

Attribute	Specification
Purpose	Revenue overview.
Access	Provider / Driver / Houser
Entry point	Dashboard tab
Layout	Weekly earnings card (% change); stat tiles (Trips/Services/Rating); today's bookings.
Firebase	Aggregate bookings/rides (Cloud Function)

6.58 Edit / Delete Service [N]

Attribute	Specification
Purpose	Edit/remove a service.
Access	Service Provider
Entry point	Service Listings → tap
Layout	Pre-filled form; Save; Delete.
Firebase	Update/delete services/{id}

6.59 List Animal for Sale/Adoption [N]

Attribute	Specification
Purpose	Create animal listing.
Access	Seller
Entry point	Place Pets For Sale
Layout	Photos; details; price or Free Adoption toggle; description; Submit.
States	Empty, Validation, Saving, Success
Firebase	Create animalListings; photos → Storage

6.60 Manage Listings [N]

Attribute	Specification
Purpose	Manage own animal listings.
Access	Seller
Entry point	Seller area
Layout	Listing cards (status); Edit/Delete; Mark Sold; + FAB.
Firebase	CRUD animalListings where sellerId == uid

7. Admin Panel Specifications

The admin app is a mobile app (Flutter, same codebase/shared components as the consumer app where practical), accessible only to accounts with role admin. It uses the Firebase Admin SDK via Cloud Functions for privileged operations and connects to the SQL database for ride/supply-request allocation. Authentication is Firebase Auth with a custom admin claim.

7.1 Admin Pages

Page	Purpose	Key Actions
Dashboard	Platform overview at a glance	View counts: users, pending ride requests, pending verifications, today's revenue, open rescue pins
Pending Driver Requests	View ride requests that need fulfilment	Approve request / change status to 'driver allocated' or 'completed'
Pending Medication/Supply Requests	View user requests for supplies that need fulfilment	Change status to 'approved', 'completed', or 'unavailable'
User Management	View and manage all users	Search/filter users; view profile; suspend/reactivate; change role; view user's orders
Verification Queue	Approve role/seller verifications	View submitted documents; Approve / Reject with reason; set verified badge
Listings Moderation	Moderate animal listings	View flagged listings; remove listing; warn seller
Clinic Management	Maintain clinic directory	Add/edit clinic; set is24Hours; set hours, location, phone
Chat Moderation	Review flagged chat threads	Open thread (read-only); close thread; warn/suspend user
Rescue Oversight	Monitor rescue pins	View all pins; close stale/abusive pins
Reports	Operational and financial reports	Generate and export (see Section 12)
Settings	Platform configuration	Fare rates, commission %, payment toggles
Audit Logs	Trace privileged actions	View admin action log; filter by admin/date/action
Notifications	Broadcast messages	Send push/in-app announcement to a segment

7.2 Admin Access Control

- Only Firebase Auth accounts with a custom claim admin == true can access the panel
- All write operations go through Cloud Functions that re-check the admin claim server-side
- Every privileged action writes an entry to auditLogs (who, what, when, target)

Security: Admin privileges are granted only by setting a custom claim via a secured Cloud Function or the Firebase console — never from the client app.

8. Database Design

A managed SQL database is the main datastore for core transactional/relational data (users, pets, rides, bookings, orders, payments, drivers roster). Firestore is used as secondary storage where its model fits better — flexible listing content, chat, and the rescue map. All tables/collections, key fields, and relationships are below. Field types: S=string, N=number, B=boolean, T=timestamp, G=GeoPoint (Firestore only), M=map, A=array, R=reference/id/foreign key.

users *all accounts*

Field	Type	Req	Notes
uid	S	Y	Firebase Auth UID (doc id)
username	S	Y	3–30 chars
phone	S	Y	E.164
email	S	N	optional
city	S	N	Riyadh / Cairo / ...
role	S	Y	pet_owner seller provider houser clinic admin
roles	A	N	array if multiple roles held
verified	B	Y	true after approval (sellers/providers)
avatarUrl	S	N	Storage URL
createdAt	T	Y	server timestamp
suspended	B	N	admin can set

pets *user-owned pets*

Field	Type	Req	Notes
id	S	Y	doc id
ownerId	R	Y	→ users.uid
name	S	Y	
species	S	Y	cat/dog/bird/...
breed	S	N	
age	N	N	years
gender	S	N	male/female
photoUrl	S	N	Storage URL

rides *transport requests (SQL)*

Field	Type	Req	Notes
id	S	Y	row id
userId	R	Y	→ users.uid
driverId	R	N	null until allocated by admin
petId	R	Y	→ pets.id
status	S	Y	requested driver_allocated in_progress delivered cancelled
pickup	S	Y	address/description
destination	S	Y	address/description
carType	S	Y	any specific
fareAmount	N	N	set by Cloud Function only
paymentStatus	S	Y	pending paid refunded
paymentTxnId	S	N	gateway id
allocatedBy	R	N	→ admin uid who allocated the driver

Field	Type	Req	Notes
cancellationReason	S	N	
createdAt / allocatedAt / completedAt	T	Y/N	lifecycle timestamps

drivers *driver roster (SQL — admin-managed, not app accounts)*

Field	Type	Req	Notes
id	S	Y	row id
name	S	Y	
phone	S	Y	
vehicleType	S	Y	
plate	S	Y	
active	B	Y	available for allocation
rating	N	N	avg, optional

services / bookings *service catalog + bookings*

Field	Type	Req	Notes
service.providerId	R	Y	→ users.uid
service.type	S	Y	grooming/treatment/hiking/showering
service.price / duration	N	Y	
booking.userId / providerId / serviceId / petId	R	Y	references
booking.slot	T	Y	reserved slot
booking.status	S	Y	pending confirmed completed cancelled
booking.paymentStatus	S	Y	pending paid refunded

housingListings / housingBookings *boarding*

Field	Type	Req	Notes
listing.houserId	R	Y	→ users.uid
listing.type	S	Y	home clinic med24
listing.pricePerNight	N	Y	
listing.acceptedAnimals	A	Y	
listing.location	G	Y	
booking.dates	A	Y	blocked date range
booking.status / paymentStatus	S	Y	

animalListings *buy/sell/adopt*

Field	Type	Req	Notes
id	S	Y	doc id
sellerId	R	Y	→ users.uid
species / breed / age / gender	S/N	Y	
price	N	N	null = adoption
photos	A	Y	Storage URLs
verified	B	Y	admin-set; licensed tier
status	S	Y	active sold

supplyRequests *medical-supply requests (manually fulfilled by admin)*

Field	Type	Req	Notes
requesterId	R	Y	→ users.uid
title / category / description	S	Y	
photoUrl	S	N	optional
status	S	Y	pending approved completed unavailable
reviewedBy	R	N	→ admin uid

clinics / clinicBookings *vet directory + emergency booking*

Field	Type	Req	Notes
clinic.name / address / phone	S	Y	admin-maintained
clinic.location	G	Y	
clinic.is24Hours	B	Y	admin-set
clinic.hours	M	Y	per-day open/close
booking.userId / clinicId	R	Y	
booking.paymentStatus	S	Y	

rescuePins *community rescue*

Field	Type	Req	Notes
id	S	Y	doc id
reporterId	R	Y	→ users.uid
animalType	S	Y	cat/dog/bird/other
urgency	S	Y	critical needs_help stable
description	S	N	≤200 chars, filtered
location	G	Y	
geohash	S	Y	nearby query
status	S	Y	open closed
responders	A	N	uids who offered help
createdAt	T	Y	auto-expire after period

chatThreads / messages *controlled chat*

Field	Type	Req	Notes
thread.participants	A	Y	[buyerUid, sellerUid]
thread.context	M	Y	{type, refId} of transaction
thread.locked	B	Y	true when txn done
message.senderId	R	Y	
message.text	S	Y	validated (no phone/email/url)
message.createdAt	T	Y	immutable

verificationRequests / auditLogs / paymentMethods / notifications *supporting*

Field	Type	Req	Notes
verif.userId / role / documents[]	R/A	Y	status pending approved rejected, reason
audit.adminId / action / targetId / at	R/S/T	Y	privileged-action trail
paymentMethod.userId / type / token	R/S	Y	tokenized; no PAN stored
notification.userId / type / body / read	R/S/B	Y	in-app inbox

8.1 Indexes

- SQL: rides (status, createdAt), (driverId, status), (userId, createdAt); bookings (providerId, status), (userId, createdAt); housingBookings (listingId, dates) for overlap checks
- Firestore: animalListings composite (status, species, createdAt); rescuePins (status, geohash)
- clinics: (is24Hours) — Firestore or SQL depending on final placement; treated as admin-maintained reference data either way

8.2 Data Retention

- Completed rides/bookings/orders retained indefinitely for the owner's history (no PII beyond account)
- Rescue pins auto-close after a configurable window (e.g. 72h) but retained for analytics
- Chat messages retained while the account exists; deletable on account deletion request
- Verification documents retained only until decision + a grace window, then deletable

9. Firebase Architecture

Service	Used For
Authentication	Phone (OTP) + Google sign-in; custom admin claim
SQL Database (managed)	Main storage for users, pets, rides, bookings, orders, drivers roster, payments
Firestore	Secondary storage: listings content, chat, rescue pins
Storage	Pet photos, listing images, verification documents
Cloud Functions	Fare calc, payments, booking transactions, chat validation, notifications, admin operations (driver/supply allocation), scheduled cleanups
Hosting	Static assets (if needed); admin is a mobile app, not web
Cloud Messaging (FCM)	Push notifications (ride/booking status, chat, verification)
Analytics	User/event analytics
Crashlytics	Mobile crash reporting
Security Rules	Firestore + Storage access enforcement; SQL access enforced in Cloud Functions

9.1 Key Cloud Functions

Function	Trigger	Responsibility
onUserCreate	Auth create	Create user record with default role
calculateFare	Callable	Compute fare from distance/time/surge (authoritative)
allocateDriver	Callable (admin)	Admin allocates a driver to a pending ride; sets status driver_allocated; notifies user
processPayment	Callable	Charge via gateway; idempotent; update payment status
processRefund	Triggered (ride/booking cancelled)	Refund via gateway; update + notify
bookSlot / bookHousing	Callable	Transaction to prevent double-booking
validateMessage	Firestore (message create) / callable	Reject phone/email/url; allow or delete
approveVerification	Callable (admin)	Set verified; set badge; audit log; notify
resolveSupplyRequest	Callable (admin)	Set supply request status (approved/completed/unavailable); notify
closeStalePins	Scheduled	Close rescue pins older than window
aggregateEarnings	Callable / scheduled	Compute provider earnings for dashboard

9.2 Security Rules Principles

- Default deny: nothing is readable/writable unless a rule allows it
- Ownership checks: request.auth.uid must match the resource owner field (Firestore) / requester id (SQL, enforced in Cloud Functions)
- Role checks: sensitive writes (verified, fareAmount, is24Hours, driver allocation) only via Admin SDK in Cloud Functions
- Storage rules mirror ownership; documents folder restricted to owner + admin

```
// Example Firestore rule (users) match /users/{uid} {
  allow read: if request.auth.uid == uid || isAdmin();
  allow create: if request.auth.uid == uid;
  allow update: if request.auth.uid == uid && !("verified" in request.resource.data.diff(resource.data).affectedKeys());
}
function isAdmin() {
  return request.auth.token.admin == true;
}
```

10. Notification System

All notifications are delivered via Firebase Cloud Messaging (push) and recorded in the notifications collection (in-app inbox). SMS is used only for OTP via Firebase Phone Auth.

Trigger Event	Recipient	Push	In-App	SMS
OTP requested	User	—	—	Yes
First login / welcome	User	—	Yes	—
Driver allocated (admin action)	User	Yes	Yes	—
Ride status change / delivered	User	Yes	Yes	—
New service/housing booking	Provider / Houser	Yes	Yes	—
Booking confirmed / reminder	User	Yes	Yes	—
New chat message (active thread)	Recipient	Yes	Yes	—
Listing interest flagged	Seller	Yes	Yes	—
Purchase / order placed	Seller	Yes	Yes	—
Verification approved / rejected	Applicant	Yes	Yes	—
New Critical rescue pin nearby	Nearby users	Yes	Yes	—
Someone offered to help (rescue)	Reporter	Yes	Yes	—
Pay-to-book confirmed / reminder	User	Yes	Yes	—
Refund processed	User	Yes	Yes	—
Admin broadcast	Segment	Yes	Yes	—

11. Security Requirements

11.1 Application Security

- All network traffic over HTTPS/TLS (enforced by Firebase)
- No secrets, API keys, or gateway credentials in the client app; stored in Cloud Functions config / Secret Manager
- Payment card data never stored; only tokenized references from the gateway
- OTP brute-force protection: max 3 requests / 10 minutes / number; lockout on repeated failure
- Input validation on every form (client) and re-validation in Cloud Functions (server)

11.2 Firebase Security

- Default-deny Security Rules for Firestore and Storage
- Ownership- and role-based access enforced in rules; privileged writes only via Admin SDK
- Driver-assignment uses a transaction to prevent race conditions / double assignment
- Payment writes (fareAmount, paymentStatus) only by Cloud Functions, never the client
- Admin actions gated by custom claim and logged to auditLogs

11.3 Backup Strategy

- Scheduled Firestore export to a Cloud Storage bucket (daily) — owned by the client
- Storage bucket versioning enabled for uploaded media
- Documented restore procedure in the handover documentation

11.4 Error Handling, Logging, Monitoring

- All Cloud Functions wrap logic in try/catch and return structured errors; failures logged
- Client surfaces user-friendly error states (see per-screen states in Section 6)
- Crashlytics for mobile crash reporting; Cloud Functions logs for backend
- Analytics events for key funnels (sign-up, ride request, booking, payment)

12. Reporting & Analytics

Reports are generated in the admin app via Cloud Functions aggregating data from SQL and Firestore. Exports available as CSV.

Report	Contents	Filters
Revenue Report	Total revenue by stream (transport, services, housing, supplies, clinic) and commission earned	Date range, city, stream
Rides Report	Ride count, completion rate, cancellation rate, avg fare	Date range, city, driver
Bookings Report	Service/housing bookings, completion, no-shows	Date range, provider
User Growth	New sign-ups, active users, by role	Date range, role
Verification Report	Pending / approved / rejected counts, avg approval time	Date range, role
Rescue Activity	Pins created, resolved, response rate	Date range, city, urgency
Driver Performance	Trips, ratings, earnings, cancellations per driver	Date range, driver

- In-app analytics via Firebase Analytics; key events: sign_up, ride_requested, ride_completed, booking_created, payment_success, payment_failure, rescue_pin_created

13. Project Phases (10 Weeks, 3 Phases)

Note: Phase boundaries are deliberately sequenced so the core transactional flow (auth → pet → ride → pay) is proven first, then expanded. Each phase ends with testing, a client demo, and acceptance sign-off before the next begins.

Phase 1 — Foundation & Core Transport (Weeks 1–4)

Features

- Project setup: Flutter app skeleton, Firebase project, Git repo, CI basics, folder architecture, design-system tokens
- Authentication (login, sign-up, OTP, forgot password, Google)
- Account-type selection + document upload + verification request
- User profile + My Pets (CRUD)
- Pet transport end-to-end: ride request, fare preview, admin driver allocation, status, delivery confirmation
- Payment integration (transport) + payment success/failure
- Push notifications for ride lifecycle

Pages

- Screens 1–14, 24–26 (auth, profile, pets, home, sidebar, checkout), 34–40 (full ride flow), 53–55 (account/verification)

Deliverables

- Working auth + profile + pets, working ride request + admin allocation + payment, Arabic/English on all Phase-1 screens
- SQL schema for users, pets, rides, drivers roster, paymentMethods; Firestore schema + Security Rules for verificationRequests
- Cloud Functions: onUserCreate, calculateFare, allocateDriver, processPayment

Acceptance Criteria

- A new user can register, verify OTP, add a pet, request a ride, have a driver allocated by admin, pay, and track to delivery with no errors
- Fare is computed server-side and cannot be altered by the client
- Fare is computed server-side and cannot be altered by the client
- All Phase-1 screens render correctly in both RTL and LTR

Tasks

Type	Tasks
Development	App scaffold, auth, profile, pets, ride flow, payment, FCM, Phase-1 Cloud Functions, rules
Testing	Unit (fare, validation), integration (auth, ride accept transaction), E2E (book→pay→deliver), RTL checks
Deployment	Firebase project live, Functions deployed, test builds (TestFlight / internal APK)

Phase 2 — Services, Housing & Marketplace (Weeks 5–8)

Features

- Pet-care services: provider discovery, profile, calendar/slot booking, payment, confirmation
- Pet housing: search, detail, availability, booking, payment
- Buy/Sell/Adopt animals: listings, detail, filters, verified seller tier, create/manage listings
- Pet medical supplies: request submission, status tracking, admin manual fulfillment
- Controlled chat: history + chat screen with message validation and auto-lock
- Provider/seller management: appointments, service listings, manage animal listings, earnings dashboard

Pages

- Screens 15–23 (chat, listings, adoption, verified, seller/provider details), 27–33 (services), 46–51 (housing + supplies), 55–60 (provider/seller management)

Deliverables

- Working services, housing, marketplace, supplies requests, and chat; provider/seller dashboards
- Firestore/SQL schemas + rules for services, bookings, housingListings, housingBookings, animalListings, supplyRequests, chatThreads/messages, reviews
- Cloud Functions: bookSlot, bookHousing, validateMessage, processRefund, aggregateEarnings

Acceptance Criteria

- A user can book a service for a specific slot with no double-booking, pay, and receive confirmation
- A seller can create, edit, mark-sold, and delete listings; verified flag is admin-only
- Chat blocks phone numbers, emails, and URLs and auto-locks when the transaction completes
- All Phase-2 screens render in RTL and LTR

Tasks

Type	Tasks
Development	Services, housing, marketplace, supplies, chat, dashboards, Phase-2 Functions, rules
Testing	Unit (slot transaction, message filter), integration (booking, refund), E2E (book service, buy animal, chat), RTL checks
Deployment	Functions deployed, updated test builds, regression against Phase 1

Phase 3 — Emergency, Rescue, Admin & Handover (Weeks 9–10)

Features

- Emergency 24/7 clinic finder + clinic detail + pay-to-book
- Rescue map + drop pin + 'I can help' + nearby notifications
- Mobile admin app: dashboard, pending driver/supply requests, user mgmt, verification queue, listings/chat moderation, clinic mgmt, reports, settings, audit logs, broadcast
- Analytics + Crashlytics wired; final hardening
- Full documentation, testing sign-off, and complete handover

Pages

- Screens 41–45 (clinic finder, detail, pay-to-book, rescue map, drop pin) + full admin app (Section 7)

Deliverables

- Working clinic finder, pay-to-book, rescue map, and admin app
- Firestore/SQL schemas + rules for clinics, clinicBookings, rescuePins, auditLogs, notifications
- Cloud Functions: approveVerification, resolveSupplyRequest, closeStalePins, report aggregations
- All handover items in Section 16; ownership transfer complete (Section 17)

Acceptance Criteria

- A user can find a 24/7 clinic, pay-to-book, and chat; a user can drop a rescue pin and others get notified
- Admin can approve a verification, which sets the verified badge and writes an audit log
- All 60 screens complete in RTL and LTR; all credentials and Firebase/SQL ownership transferred

Tasks

Type	Tasks
Development	Clinic finder, pay-to-book, rescue, admin app, analytics/crashlytics, hardening
Testing	Full regression across all phases, security testing, permission testing, UAT, performance pass
Deployment	Production Firebase + SQL, admin app store-ready build, store-ready consumer builds, handover

14. Testing Requirements

No feature is complete until tested. The developer creates and executes a test plan covering every area below, and delivers the testing artifacts in Section 16.

14.1 Test Types & Coverage

Test Type	Applied To
Unit	Fare calculation, validation rules, message filtering, slot/date conflict logic, earnings aggregation
Integration	Auth flows, driver allocation, payment + booking creation, refund, verification approval
End-to-End	Register→pet→ride→pay→deliver; book service; buy/adopt animal; housing booking; pay-to-book; rescue pin; chat
User Acceptance (UAT)	Client validates each phase against acceptance criteria before sign-off
Regression	Re-run prior-phase suites at the start of each new phase and before handover
Security	Security Rules (allow/deny), privileged-write attempts, auth bypass attempts
Permission	Each role can access only permitted pages/actions (matrix in Section 4.1)
Performance	Listing pagination, map with many pins, cold-start times

14.2 Mandatory Test Areas

- Authentication, registration, login, password reset, permissions, role management
- CRUD on pets, listings, services; search; filters
- Bookings, ride lifecycle, payments, refunds
- Notifications, file uploads, reports, analytics
- Admin functionality, Security Rules, Cloud Functions, payment-gateway integration
- Mobile / tablet responsiveness; Arabic RTL and English LTR; error handling; edge cases; validation rules

14.3 Firebase Emulator (Required)

Mandatory: The Firebase Emulator Suite must be configured from the start of Phase 1. All Cloud Functions, Security Rules, and Firestore interactions must be testable locally against the emulator before deploying to production. No payment or ride function may be written without a corresponding emulator-based test.

14.4 Definition of Done

A feature is complete only when ALL of the following are true:

- **Development complete**
- **Code reviewed**
- **Tests written and passing**
- **Security validation complete**
- **Documentation updated**
- **Feature demonstrated to client**
- **Client accepts the functionality**

15. Acceptance Criteria (Consolidated)

Final project acceptance requires all of the following:

- All 60 screens implemented and functional in both Arabic RTL and English LTR
- Core transport flow works end-to-end: request, admin driver allocation, status, delivery, server-side fare
- Services, housing, marketplace, supply requests, clinic finder, pay-to-book, rescue map, and chat all functional
- Controlled chat blocks contact info and auto-locks; no media sharing
- Payment and refund flows work via the gateway with idempotency
- Admin app functional with driver/supply-request allocation, verification approvals, moderation, reports, settings, and audit logs
- Security Rules and Cloud Function checks enforce role/ownership; privileged writes only server-side
- All testing artifacts delivered and sign-off completed
- All documentation delivered; full handover and ownership transfer completed

16. Deliverables

16.1 Source & Configuration

- Complete Flutter source code (frontend)
- All Cloud Functions source code (backend)
- Security Rules (Firestore + Storage) and SQL access-control logic in Cloud Functions
- Configuration files and environment definitions
- Database schema definitions (SQL tables + Firestore collections + indexes)
- Deployment scripts / commands
- Completed Figma source file (all 60 screens, components, styles)

16.2 Documentation

- Architecture documentation (app structure, data flow, Firebase architecture)
- Database schema documentation
- Deployment documentation
- Environment setup guide (run locally + emulator)
- Environment-variables documentation
- Admin guide
- User guide
- API / Cloud Functions documentation
- Handover documentation

16.3 Testing Artifacts

- Test plan, test cases, test execution results
- Bug tracking log and bug resolution log
- Final testing sign-off

16.4 Handover & Access Transfer

- Git repository ownership transferred to the client
- All Firebase resources and the SQL database instance owned by the client's account
- All credentials, API keys, environments, and deployment access transferred to the client
- Knowledge-transfer session(s) so the client (or any future developer) can maintain the system

Source-code quality: Code must be fully editable, readable, transferable, organized, production-ready, and free from vendor lock-in where practical. Business logic must be separated from UI; folders and files must have clear purpose; architecture must be documented.

17. Ownership & Intellectual Property

The following terms govern ownership and are a condition of the engagement.

- **The client owns 100% of all code, designs, database structure, documentation, and assets — current and future.**
- **No ownership of any kind remains with the developer.**
- **There is no revenue share.**
- **There is no equity.**
- **There are no partnership rights.**
- **There are no licensing restrictions that would prevent the client from freely modifying the project.**
- **The client may hire any future developer to maintain or extend the project.**
- **The developer may not reuse the proprietary business logic, designs, or internal project assets without the client's written permission.**

17.1 Engagement Status

- The developer is engaged strictly as an independent contractor
- The developer is not a co-founder, partner, shareholder, stakeholder, or owner
- All source code must be delivered, and a full project handover is required at completion
- The codebase must be maintained in a Git repository owned by the client; all Firebase resources and the SQL database must be owned by the client

17.2 Maintainability Independence

The final system must be maintainable by any competent developer without the original developer. The client must not depend on the original developer for: bug fixes, feature additions, deployments, infrastructure access, database access, or Firebase access. Knowledge transfer and complete handover are mandatory deliverables.

18. Risks & Assumptions

18.1 Assumptions

- Existing 25 Figma screens' status is to be re-verified; all 60 screens should be treated as requiring design/build confirmation
- Client provides a payment-gateway merchant account (Moyasar/Tap) and any required business credentials
- Client provides a Google Maps / Places API key (billing enabled) for maps, geocoding, and directions
- Single market for this engagement: Saudi Arabia
- Clinic 24/7 data is provided/verified by the client/admin, not scraped automatically
- Rescue map intentionally omits photo upload in this scope to control storage cost
- Drivers are coordinated manually by admin and are not app accounts in this MVP; driver roster is maintained directly in the SQL database by admin

18.2 Risks

Risk	Impact	Mitigation
Scope is large for 10 weeks / fixed price	Timeline pressure	Strict phase gates; out-of-scope items deferred to V2
Payment-gateway integration delays	Blocks Phase 1 payments	Confirm merchant account before Phase 1; use emulator + sandbox first
RTL defects discovered late	Rework across screens	RTL tested per screen each phase, not at the end
Double-booking (slot/date)	Data integrity	Transactions for slot/date booking
Inadequate testing of Cloud Functions	Production payment/ride bugs	Mandatory Firebase Emulator + tests (Section 14.3)
SQL + Firestore dual-storage drift	Data inconsistency	Single source of truth per entity defined in Section 8; no entity split across both stores
Key-person dependency on developer	Maintenance risk	Mandatory documentation + handover (Sections 16–17)

19. Open Questions

The following require client decisions. Reasonable assumptions are noted where the project can proceed without an immediate answer.

#	Question	Working Assumption
1	Which payment gateway — Moyasar or Tap?	Assume Moyasar (Mada + card); swappable behind a payment service layer
2	Does Google Sign-In require extra onboarding screens (favorite animals, etc.)?	Assume no extra screens for MVP; collect later in profile
3	Sign-up onboarding: collect favorite animals / animal details?	Assume optional, post-registration in profile
4	Commission percentages per stream (transport, services, housing)?	Configurable in admin Settings; defaults set with client
5	Rescue-pin auto-close window (hours)?	Assume 72 hours, configurable
6	Surge-pricing rules for transport?	Assume simple time-based surge, configurable; off by default
7	Is the full admin app required within the 10 weeks or can it be minimal?	Assume core admin (driver/supply allocation, verification, moderation, reports) in Phase 3
8	Driver vetting requirements (since drivers are admin-managed, not self-registered)?	Assume admin manually verifies drivers offline; only a basic roster (name, phone, vehicle, plate) is stored
9	Refund policy timing and conditions?	Assume gateway refund on cancellation; client to confirm exact rules

20. Appendix

A. Screen Index (60)

Status: E = exists in Figma · N = new · P = partial. All statuses to be re-verified (see Section 1). Full specs in Section 6.

#	Screen	Status	#	Screen	Status
1	Login	E	31	Select Payment (Service)	E
2	OTP Confirmation	E	32	Service Order Summary	N
3	Set Password	E	33	Service Order Placed	E
4	Register	E	34	Ride Request	E
5	Sign In With Google	E	35	Delivery Path Review	E
6	Forgot Password	N	36	Setup Payment (Ride)	E
7	Display User Info	N	37	Searching For a Driver	E
8	Edit User Info	N	38	Wait For Driver	E
9	My Pets List	N	39	Ride Status	N
10	Add Pet	N	40	Delivery Confirmation	E
11	Edit Pet Info	N	41	24/7 Clinic Finder	N
12	Service History	N	42	Clinic Detail	N
13	Home Page	E	43	Pay-to-Book Emergency	N
14	Side Bar	E	44	Rescue Map	N
15	Chat History	N	45	Drop a Rescue Pin	N
16	Chat Screen	N	46	Housing Search/Listing	N
17	Search / Listing	E	47	Housing Detail	N
18	Filter Page	E	48	Housing Booking Flow	N
19	Animal Detail	E	49	Supplies Request/Search	N
20	Ask About Pet	E	50	Supplies Request Detail	N
21	Seller/Provider Details	E	51	Create Supplies Request	N
22	Verified Sellers Section	N	52	Account Type Selection	P
23	Adoption Listing	N	53	Document Upload	E
24	Checkout — Addr & Pay	N	54	Verification Status	N
25	Payment Success	N	55	Available Appointments	N
26	Payment Failure	N	56	Service Listings Mgmt	N
27	Search Service Provider	E	57	Earnings / Dashboard	N
28	Setup Animal (Service)	E	58	Edit / Delete Service	N
29	Service Provider Profile	E	59	List Animal Sale/Adopt	N
30	Calendar / Time Slot	P	60	Manage Listings	N

Note: Ride Safety Compliance (previously screen 39) is removed in this revision — no driver-side app, so no driver-facing handoff screen. Screen count reduced from 61 to 60.

B. Technology Summary

Layer	Choice
Mobile framework	Flutter (iOS + Android)
State management	Developer's choice (Riverpod or BLoC recommended), documented

Layer	Choice
Backend	Managed SQL database (main) + Firebase: Auth, Firestore (secondary), Storage, Functions, FCM, Analytics, Crashlytics
Maps	Google Maps / Places (client-provided key)
Payments	Moyasar or Tap (Mada + card), behind a payment service layer
Localization	Flutter l10n (.arb) + RTL via Directionality / EdgeInsetsDirectional
Admin app	Mobile (Flutter), Admin SDK + SQL access via Cloud Functions

C. Glossary

- OTP — One-Time Password sent by SMS for phone verification
- RTL / LTR — Right-to-Left (Arabic) / Left-to-Right (English) layout direction
- FCM — Firebase Cloud Messaging (push notifications)
- SQL — the managed relational database serving as main storage for core transactional data (Section 8)
- Geohash — encoded location string enabling efficient nearby queries (used for Firestore-stored rescue pins)
- Idempotency — ensuring a repeated payment request does not double-charge